Exam Date & Time: 17-Sep-2025 (10:45 AM - 12:15 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

Discrete Mathematical Structures [MAT 2101]

Marks: 30 **Duration: 90 mins.**

MCQ

Answer all the questions. Section Duration: 20 mins

1) Which ONE of the following is a monoid but not a group...

1)	The set of all non-singular real matrices of order 2 under "matrix addition".		2)	The set of all integers under "subtraction".		3)	The set of all subsets of a non-empty set under the set operation "union of sets".		4)	The set of all real square matrices of order 2 under "matrix addition".	

(0.5)

Correct option is: 3

2) Which ONE of the following is NOT a subgroup of a group $\{1, -1, i, -i\}$ with respect to multiplication?

1)	$\{i, -i\}$		2)	$\{1\}$		3)	$\{1, -1\}$		4)	$\{1, -1, i, -i\}$	

(0.5)

Correct option is: 1

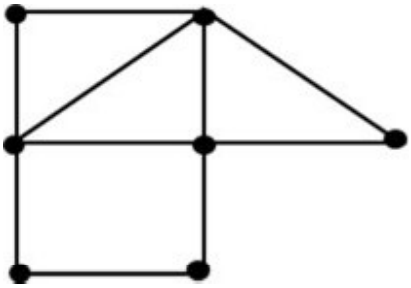
3) Singular values of the matrix $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ are...

1)	1, 9		2)	$0, \sqrt{3}$		3)	1, 3		4)	$1, \sqrt{3}$	

(0.5)

Correct option is: 3

4) Radius of the following graph is... (0.5)



1)	2		2)	3		3)	4		4)	7	

Correct option is: 1

5) Given a graph G = (V, E), the square graph G^2 is defined as the graph with the same vertex set $V(G^2) = V(G)$, where two vertices u, v ∈ V are adjacent in G^2 whenever d(u, v) ≤ 2 in G. If G = C_4 (cycle of length 4), then G^2 is isomorphic to...

1)	C_4		2)	K_4 (Complete graph with 4 vertices)		3)	$K_{2,2}$ (complete bi-partite graph)		4)	Path P_4	

(0.5)

Correct option is: 2

6) Let G={1,3,5,7} be a group under the operation "multiplication modulo 8". Then the inverses of the elements 3, 5 and 7 respectively, are

1)	3, 5 and 7		2)	5, 3 and 7		3)	5, 7 and 3		4)	7, 5 and 3	

(0.5)

Correct option is: 1

7) Let H = 3ℤ (set of all integers which are multiples of 3) and K = 5ℤ (set of all integers which are multiples of 5) be subgroups of (ℤ, +). Which ONE of the following is NOT a subgroup of ℤ?

1)	$H \cap K$		2)	$H + K = \{h + k : h \in H, k \in K\}$		3)	$H \cup K$		4)	{0}	

(0.5)

Correct option is: 3

8) If the number of closed walks of length 3 in a graph G is 24, then the number of triangles in G is…

1) 2 2) 4 3) 6 4) 8

(0.5)

Correct option is: 2

9) Let G be a graph with 21 edges. Suppose 3 vertices of G are of degree 4 and the remaining vertices are of degree 2. Then how many vertices G has? (0.5)

1)	18		2)	10		3)	16		4)	9	

Correct option is: 1

10) If $A=\begin{bmatrix}1&1&0\\1&0&1\\0&1&1\end{bmatrix}=\begin{bmatrix}1&0&0\\l_{21}&1&0\\l_{31}&l_{32}&1\end{bmatrix}\begin{bmatrix}u_{11}&u_{12}&u_{13}\\0&u_{22}&u_{23}\\0&0&u_{33}\end{bmatrix}$ then l_{21}=... (0.5)

1)	1		2)	0		3)	2		4)	-1	

Correct option is: 1

Descriptive Questions

Answer all the questions.

11) Find a singular value decomposition of the matrix $A=\begin{bmatrix}1&0&0\\0&2&0\end{bmatrix}$. (4)

12) Solve the following system of equations by using LU decomposition method.

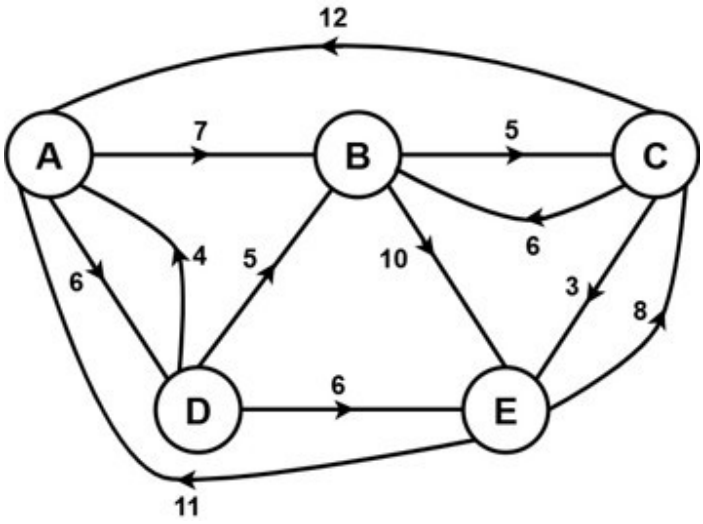
$$x+y+z=1$$

$$4x+3y-z=6$$

$$3x+5y+3z=4$$

(3)

13) Use Dijkstra's algorithm to find the shortest path from B to all other vertices of G in the following network.

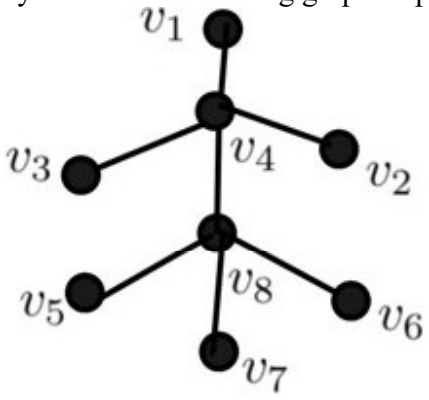


(3)

14) Let $*$ be defined on the set of integers as $x * y = x + y - 2, \forall x, y \in \mathbb{Z}$. Then verify whether $(\mathbb{Z}, *)$ is a group. Justify your answer. Is $(\mathbb{Z}, *)$ Abelian? (3)

15) i). Let (G, *) be a group and H be a nonempty subset of G. If $a * b^{-1} \in H$ for all $a, b \in H$ then show that H is a subgroup of G.
ii). Let G = $\mathbb{Z}_8$ be a group under the operation 'addition modulo 8'. Construct the cayley table for the subset H={0, 2, 4, 6} of G under 'addition modulo 8'. Is H a subgroup of G? Justify your answer. (3)

16) If G is a connected bi-partite graph with at least 3 vertices, then show that G has no odd cycles. Is the following graph bi-partite? If so, write down the partite sets of the graph.



(3)

17) Show that the number of vertices in a self-complementary graph G is of the form 4k or 4k+1, where k is a positive integer. (2)

18) Show that if G is a group and $a^2 = e$ for all $a \in G$, then G is Abelian. (2)

19) A tree T with 50 pendant vertices (Vertices of Degree 1) has an equal number of vertices of degree 2,3,4, and 5 and no vertices of degree greater than 5. Find the number of vertices in T. (2)

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EVALUATION SCHEME – MAT 2101 MID-TERM EXAMINATION (2025)

11. Find a singular value decomposition of $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}$. (4M)

Ans. $M = A^T A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. $-\,1/2$

$\lambda = 4, 1, 0 \implies \sigma = 2, 1, 0.$ $-\,1$

$V = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $-\,1$

$U = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $-\,1$

Then the SVD of A is

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad -\,1/2$$

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12. Solve the following system of equations using LU decomposition. (3M)

$$x + y + z = 1$$
$$4x + 3y - z = 6$$
$$3x + 5y + 3z = 4.$$

Ans. $A = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -5 \\ 0 & 0 & -10 \end{bmatrix} = LU$ $-\,2$

$Ly = b \implies y = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$. $-\,1/2$

$Ux = y \implies x = \begin{bmatrix} 1 \\ 1/2 \\ -1/2 \end{bmatrix}$. $-\,1/2$

13. Use Dijkstra's algorithm to find shortest paths from B to all the other vertices in the following network. (3M)

Ans.

	A	C	D	E	
$B(0)$	∞_B	5_B	∞_B	10_B	$-\,1/2$
$C(5)$	17_C	$-$	∞_B	8_C	$-\,1$
$E(8)$	17_C	$-$	∞_B	$-$	$-\,1/2$
$A(17)$	$-$	$-$	23_A	$-$	$-\,1/2$
B	17_C	5_B	23_A	8_C	$-\,1/2$

Half a mark can be awarded for the weight matrix if it is given, but no marks should be deducted for not giving it.

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14. Let $*$ be defined on $\mathbb{Z}$ as $x * y = x + y - 2$ for all $x, y \in \mathbb{Z}$. Then verify whether $(\mathbb{Z}, *)$ is a group. Justify your answer. Is $(\mathbb{Z}, *)$ Abelian? (3M)

Ans. If $x, y \in \mathbb{Z}$, then $x * y = x + y - 2 \in \mathbb{Z}$. Hence $*$ is a binary operation on G. $-\,1/2$

Associativity of $*$:
$x * (y * z) = x + (y + z - 2) - 2 = x + y + z - 4 = (x + y - 2) + z - 2 = (x * y) * z$. $-\,1$

Existence of identity element:
Since $x * 2 = 2 * x = x + 2 - 2 = x$, and $2 \in \mathbb{Z}$, the identity element (2) exists in the group. $-\,1/2$

Existence of inverses:
$x * (4 - x) = (4 - x) * x = x + (4 - x) - 2 = 2$, and $x \in \mathbb{Z} \implies 4 - x \in \mathbb{Z}$. Hence inverses exist. $-\,1/2$

Since $x * y = x + y - 2 = y + x - 2 = y * x$, the group is Abelian. $-\,1/2$

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15. (i) Let $(G, *)$ be a group and H a nonempty subset of G. If $a * b^{-1} \in H$ for all $a, b \in G$, then show that $H \leq G$. (3M)

(ii) Consider the group $G = \mathbb{Z}_8$ under addition modulo 8. Construct the Cayley table for the subset $H = \{0, 2, 4, 6\}$ under addition modulo 8. Is H a subgroup of G? Justify your answer.

Ans. (i) Since $H \neq \emptyset$, there exists $a \in H$. Then $a, a \in H \implies e = aa^{-1} \in H$. $-\,1/2$

Now if $a \in H$, then $e, a \in H \implies ea^{-1} = a^{-1} \in H$. $-\,1/2$

If $a, b \in H$, then, $b^{-1} \in H$. Thus, $a, b^{-1} \in H \implies a(b^{-1})^{-1} \in H$. $-\,1/2$

(ii) The Cayley table of $\{0, 2, 4, 6\}$ is shown below.

$+_8$	0	2	4	6
0	0	2	4	6
2	2	4	6	0
4	4	6	0	2
6	6	0	2	4

$-\,1$

Yes, $H = \{0, 2, 4, 6\}$ is a subgroup of $\mathbb{Z}_8$. The Cayley table shows that H is closed under $+_8$, and clearly $0 \in H$. As every row of the table contains 0, H has inverses as well. $-\,1/2$

16. If G is a connected bipartite graph with at least 3 vertices, then show that G has no odd cycles. Is the following graph bipartite? If so, write down the partite sets of the graph. (3M)

Ans. Let (V_1, V_2) be a bipartition of G, and let $v_1, v_2, \ldots, v_k$ be a cycle of length k in G. $-\frac{1}{2}$

Without loss of generality, suppose that $v_1 \in V_1$. Then, since $v_2 \sim v_1$, $v_2 \in V_2$, and hence $v_3 \in V_1$, as $v_2 \sim v_3$. $-\frac{1}{2}$

Proceeding similarly, we see $v_i \in V_1$ if i is odd and $v_i \in V_2$ if i is even. $-\frac{1}{2}$

But $v_k \sim v_1$ and $v_1 \in V_1$ implies that $v_k \in V_2$. Therefore, k is even. $-\frac{1}{2}$

Yes, the given graph is bipartite. $-\frac{1}{2}$

The partite sets are $\{v_1, v_2, v_3, v_8\}$ and $\{v_4, v_5, v_6, v_7\}$. $-\frac{1}{2}$

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17. Show that the number of vertices in a self-complementary graph G is of the form $4k$ or $4k+1$, where k is an integer. (2M)

Ans. Let G be a self-complementary graph of order n and size m.

Then $|E(\overline{G})| = \binom{n}{2} - m$. $-\frac{1}{2}$

Since $G \cong \overline{G}$, $m = \binom{n}{2} - m \implies m = \frac{1}{2}\binom{n}{2} = \frac{n(n-1)}{4}$. $-\frac{1}{2}$

Hence, $4 \mid n(n-1) \implies 4|n$ or $4 \mid n-1$. $-\frac{1}{2}$

Hence, $n = 4k$ or $n = 4k+1$ for some $k \in \mathbb{Z}$. $-\frac{1}{2}$

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18. Show that if G is a group and $a^2 = e$ for all $a \in G$, then G is Abelian. (2M)

Ans. If $a^2 = e$, then $a = a^{-1}$. $-\frac{1}{2}$

We know that for all $a, b \in G$, $(ab)^{-1} = b^{-1}a^{-1}$. -1

Hence, $ab = ba$ for all $a, b \in G$, i.e. G is Abelian. $-\frac{1}{2}$

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19. A tree with 50 pendant vertices has an equal number of vertices of degrees 2, 3, 4, and 5, and no vertices of degree greater than 5. Find the number of vertices in T. (3M)

Ans. n be the order of the tree, and let x be the number of vertices of degrees 2, 3, 4, and 5 each. $-\frac{1}{2}$

Then $n = 50 + 4x$, and hence the size of the tree is $49 + 4x$. $-\frac{1}{2}$

By the Handshaking Lemma,

$$50 \times 1 + x \times (2+3+4+5) = 2(49+4x) \implies \quad -\frac{1}{2}$$

Therefore, $x = 8 \implies n = 82$. $-\frac{1}{2}$